```python
n = 5
m = 3
allocation = [
    [0, 1, 0],
    [2, 0, 0],
    [3, 0, 2],
    [2, 1, 1],
    [0, 0, 2]
]
maximum = [
    [7, 5, 3],
    [3, 2, 2],
    [9, 0, 2],
    [2, 2, 2],
    [4, 3, 3]
]
available = [3, 3, 2]
need = []
for i in range(n):
    temp = []
    for j in range(m):
        temp.append(maximum[i][j] - allocation[i][j])
    need.append(temp)
print("Need Matrix:")
for row in need:
    print(row)
finish = [False] * n
safe_sequence = []
while len(safe_sequence) < n:
    allocated = False
    for i in range(n):
        if not finish[i]:
            if all(need[i][j] <= available[j] for j in range(m)):
```

```python
                for j in range(m):
                    available[j] += allocation[i][j]
                safe_sequence.append(i)
                finish[i] = True
                allocated = True
    if not allocated:
        break
if len(safe_sequence) == n:
    print("\nSystem is in SAFE state")
    print("Safe Sequence is:")
    for i in safe_sequence:
        print("P", i, end=" ")
else:
    print("\nSystem is NOT in safe state")
```

```
Need Matrix:
[7, 4, 3]
[1, 2, 2]
[6, 0, 0]
[0, 1, 1]
[4, 3, 1]

System is in SAFE state
Safe Sequence is:
P 1 P 3 P 4 P 0 P 2
Process finished with exit code 0
```